

Durgapur Institute of Advanced Technology & Management

Topic : Rule based and Utility based Agents with example

Department: Computer Science & Engineering

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Introduction to Intelligent Agents:-

Agents are used in AI to solve real-world problems by making decisions based on inputs.

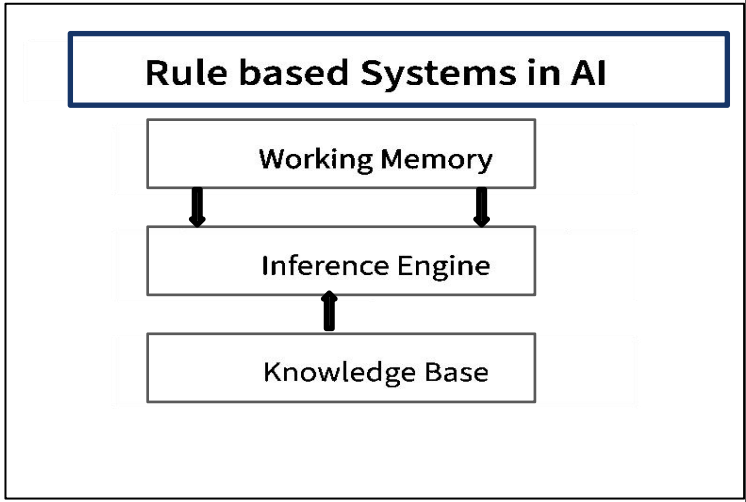
Agent Types:

(1) **Rule-Based Agents (1950-1980)**– A **rule-based agent** is an AI system that uses a set of **"if-then" rules** to make decisions.

- It checks conditions and takes actions based on those rules.
- It works well for **clear and well-defined tasks**.
- These agents are **easy to understand**, **predictable**, and **transparent**.

Example:

- IF temperature > 25°C THEN turn on fan.
- IF motion detected THEN turn on light.
- IF customer says 'hello' THEN respond with greeting.
- IF battery < 15% THEN enable power-saving mode.



➢ **Characteristics:**

- Simple, fast, easy to design.
- Deterministic and predictable.
- Not adaptive to complex or new situations

➢ **Advantages:**

- Straightforward to implement with minimal technical overhead.
- Executes quickly due to deterministic logic.
- Rule logic is transparent and easy for humans to interpret and audit.

➢ **Disadvantages:**

- Lacks flexibility in rapidly changing or unpredictable environments.
- Incapable of handling ambiguous data or probabilistic outcomes.
- Scalability becomes challenging as the number of rules grows, making the system hard to manage and prone to conflicts.

2. Utility-Based Agents (1992-2000)– A **utility-based agent** is an AI agent that makes decisions by calculating the **usefulness (utility)** of each action.It chooses the action that gives the **highest overall benefit**.Unlike goal-based agents that only aim to reach a goal, utility-based agents look for the **best** way to reach it.The **utility** can be anything — like **maximizing profit** or **saving energy**.

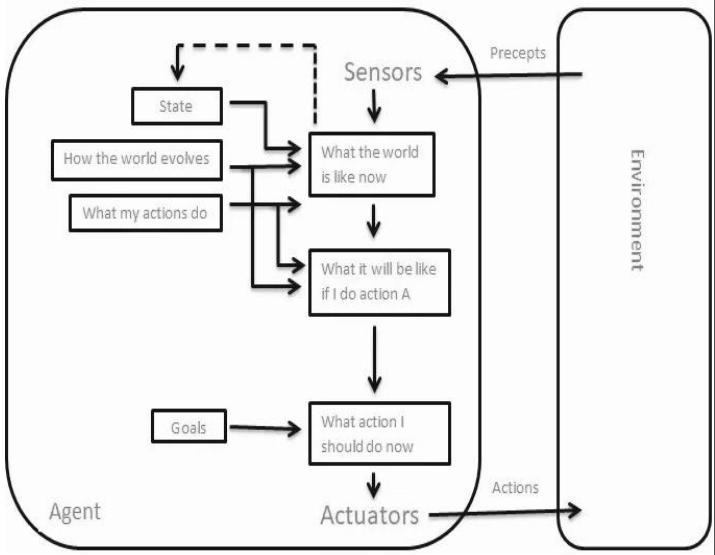
Example: Utility = (Package urgency) - (Route risk + Distance)

- Self-driving car:Utility = (Safety score * 0.6) + (Speed score * 0.3) + (Fuel efficiency * 0.1)
- Online shopping assistant:

Utility = (Product relevance * 0.5) + (User rating * 0.3) - (Price * 0.2)

- Energy management system:

Utility = (Energy savings) - (Comfort deviation + Device wear cost)



- **Characteristics:**

  •Rational and flexible.
  •Handles trade-offs and uncertainty.
  •Requires more computation and design effort.

- **Advantages:**

  •Optimizes decisions for best possible outcomes.
  •Adapts to complex, dynamic environments.
  •Can evaluate multiple goals and trade-offs.

- **Disadvantages:**

  •Complex utility functions can be difficult to design.
  •Requires more processing power and data.
  •Less transparent than rule-based systems.

➢ Visual Comparison

Feature	Rule-Based Agent	Utility-Based Agent
Decision Process	Fixed IF-THEN rules	Utility calculation
Flexibility	Low	High
Adaptability	Poor	Strong
Use of Preferences	None	Optimizes for best outcome
Example	Smart Light Switch	Smart Navigation Robot

➢ Real-Life Examples

Domain	Rule-Based Example	Utility-Based Example
Home Automation	Turn lights on at 7 PM	Adjust lights based on comfort
Healthcare	IF symptom A THEN prescribe B	Select treatment with best outcome
Gaming	NPC follows scripted path	NPC adapts to player behavior
Robotics	Avoid wall when sensor hits	Plan path minimizing risk & time

Conclusion:-

- Use **Rule-Based Agents** when:
  - The environment is predictable.
  - Rules are clearly defined.
  - Simplicity and speed are priorities.
- Use **Utility-Based Agents** when:
  - The environment is dynamic and uncertain.
  - Decisions require evaluating trade-offs.
  - Flexibility and optimization are critical.
- **Key Insight:**
  - Rule-based = simple logic, quick decisions.
  - Utility-based = adaptive intelligence, optimized decisions.
- **Choose based on:**
  - Task complexity.
  - Data availability.
  - Performance and adaptability requirements.

Thank You